mnirs v0.7.0

Muscle Near-Infrared Spectroscopy Processing and Analysis

online Shiny app

Read, process, and analyse data from muscle near-infrared spectroscopy (mNIRS) devices. Import raw data from file and return time-series data and metadata. Standardised methods for cleaning, filtering, transforming, and analysing mNIRS data. Custom plot theme and colour palette.

Workflow

- read_mnirs()
  - create_mnirs_data()
  - example_mnirs()
- resample_mnirs()
- replace_mnirs()
  - replace_invalid()
  - replace_outliers()
  - replace_missing()
- filter_mnirs()
  - filter_moving_average()
  - filter_butterworth()
- shift_mnirs()
- rescale_mnirs()
  - theme_mnirs()
  - palette_mnirs()
  - scale_colour_mnirs()
  - scale_fill_mnirs()
  - format_hmmss()
  - breaks_timespan()
- extract_intervals()
  - by_time()
  - by_label()
  - by_lap()
  - by_sample()
- plot.mnirs()

Read

`read_mnirs()` — Import mnirs data from exported files

Package-level functions

`example_mnirs()` — Paths to an included example file

`create_mnirs_data()` — Add *"mnirs"* class metadata to data

Resample

`resample_mnirs()` — Resample to a regular time grid

Clean

`replace_mnirs()` — Replace outliers, invalid, and missing data

Vector-level functions

`replace_outliers()` — Replace local outliers

`replace_invalid()` — Replace specified invalid values

`replace_missing()` — Interpolate over *NA* values

Filter

`filter_mnirs()` — Apply digital filter to mNIRS channels

Vector-level functions

`filter_butterworth()` — Apply a Butterworth digital filter

`filter_moving_average()` — Apply moving average smoothing

Transform

`shift_mnirs()` — Shift channels to a reference value

`rescale_mnirs()` — Rescale channels data range

Extract

`extract_intervals()` — Detect events and extract intervals

Interval boundary helpers

`by_time()` / `by_label()` / `by_lap()` / `by_sample()`

— Boundaries by time value, event label, lap number, or row index

Plot

`plot()` — Plot a data frame of class *"mnirs"*

`theme_mnirs()` — ggplot2 theme for mnirs plots

`palette_mnirs()` — Access the mnirs colour palette

`scale_*_mnirs()` — Colour & fill scales with mnirs palette

`format_hmmss()` — Plot axis in time units

`breaks_timespan()` — Plot axis with time breaks